

Evolution Validation Module (EVM)

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Canonical Definition

Evolution Validation Module (EVM) defines the structural conditions under which autonomous operational evolution is verified, validated, approved, and governed before modified capabilities become operational.

EVM governs evolution validation.

The module establishes the governance conditions required to ensure that every autonomous operational change satisfies predefined governance, safety, compatibility, accountability, and operational requirements before deployment.

Authorization permits evolution.

Validation confirms readiness.

EVM governs that validation.

Module Operational Role

EVM defines the evolution validation layer of AEVS.

Its role is to verify that every authorized autonomous evolution remains operationally safe, governance-compliant, and suitable for

Module Operational Space

EVM governs:

- evolution validation,
 - operational change validation,
 - governance validation,
 - autonomous capability verification,
 - deployment validation,
 - operational readiness,
 - governed evolution approval,
 - validation assurance.
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Module Function

The module applies wherever organizations must validate autonomous operational changes before operational deployment.

Minimum Implementation Framework

1. Define the Evolution Validation Object Identify which autonomous operational changes require formal validation.

2. Define Validation Conditions

Establish governance conditions governing verification, validation, testing, and deployment approval.

3. Define Validation Failure Logic

Identify conditions where autonomous evolution becomes unsafe, unverified, incompatible, incomplete, or governance- invalid.

4. Define Governance Response

Define governance actions for deployment approval, additional testing, rollback, operational restriction, escalation, or corrective improvement.

5. Preserve Validation Traceability

Preserve validation records, governance decisions, operational evidence, testing history, deployment approvals, and supporting documentation.

Use Case 1 — Autonomous Medical Diagnostic System

Scenario

An autonomous diagnostic platform receives a governance-approved improvement to its diagnostic capabilities.

Application

EVM validates the modified capability before it is introduced into clinical operation.

Result

The upgraded system enters production only after governance-approved validation.

Use Case 2 — Autonomous Manufacturing Robot

Scenario

An industrial robot receives a software enhancement improving production efficiency.

Application

EVM verifies operational safety, compatibility, and governance compliance before deployment.

Result

Operational improvements remain safe, validated, and continuously governable.

Canonical Closing Statement

Governable evolution requires governable validation. Evolution Validation ensures that autonomous operational improvements become operational only after they have been verified, validated, approved, and continuously governed.

Related Documents

→ [Autonomous Evolution Standard \(AEVS\)](#).

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Canonical Source

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